Composing layers

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every layer's inputs explicit.

Use full traces and a bug-revealing asymmetric case;separate running computation from evaluated predictions.

Keep one habit: Assess the independent formulas,preactivations,hidden values,output and declared classes with checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Trace how one **layer** supplies the next **layer input**s

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

ReLU returns 0 and 3. It transforms a numerical score. A class **prediction** requires a separate declared **output** rule.

Opening retrieval

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Look for: Trace identity ReLU and sigmoid on a numerical score

Reference example, not the only acceptable wording: Teacher explanation: Identity gives -2, 0, 2; ReLU gives 0, 0, 2. ReLU retains the positive score itself; it is not a step classifier.

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a dot product by matching feature positions

Reference example, not the only acceptable wording: Teacher explanation: Contributions are 6 and -5, so the score is 1.

Pictured end goal

By the end, I can explain composing **layer**s using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Activation functions

Dot products and weighted sums

Teach **composition** through explicit numerical traces in two 45-minute sessions. Keep the **layer** definition, order and activation visible.

The aim is to understand which values each computation receives, not to memorise a network drawing or claim successful **training**.
</details>

Outcomes

Trace how one **layer** supplies the next **layer input**s

Check that connected **layer** dimensions and feature order agree

Explain why intermediate activations are not automatically class **prediction**s

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach the source of every intermediate value

> Main,pictured and independent networks use different formulas.

Main explanation,complete flow and trace

A network connects **layer**s so that the **output** vector of one **layer** becomes the **input** vector of the next. Use a fictional numerical **input** x = [x1, x2].

The first **layer** computes z1 = x1 - x2 and z2 = x1 + x2 - 2, then applies ReLU separately: h1 = max(0, z1), h2 = max(0, z2). Its **output** is the ordered vector h = [h1, h2].

The next **layer** computes y = 2h1 - h2 + 1. It receives h, not the original x. It uses weights [2, -1] and bias 1, with identity as its **output** activation.

This is a two-**layer** computation when counting the **layer**s with weights; an **input** placeholder is not counted here. All parameters are supplied, so this lesson performs no **training**.

The two first-**layer** units both receive the same original x. They can be calculated in either order. Only after both activations are available do we evaluate the **output layer**.

Take x = [3, 1]. First compute z1 = 3 - 1 = 2 and z2 = 3 + 1 - 2 = 2. ReLU gives h = [2, 2]. The **output layer** receives exactly this ordered pair and returns y = 2 times 2 - 2 + 1 = 3.

Record z, h and y separately so that a correct final number cannot hide an incorrect intermediate step.

The code assumes numeric **input**s with the agreed meaning. Its length check does not validate units or field order. Python indices 0 and 1 correspond to the first and second hidden components.

There is no feedback from y into h during this **forward calculation**.

Trace the evidence

STAGE	EVIDENCE	RESULT
Input x=[x1,x2]	Both units receive x	z=[x1−x2,x1+x2−2]
Two affine scores z	Apply ReLU to each	h=[h1,h2]
Ordered hidden pair h	Output weights[2,−1],bias1	One output[y]with identity

Main model input x · z1=x1−x2 · z2=x1+x2−2 · h=[ReLU(z1), ReLU(z2)] · y=2h1−h2+1

> **Main model input x:** [3, 1]
>
> **z1=x1−x2:** 2
>
> **z2=x1+x2−2:** 2
>
> **h=[ReLU(z1), ReLU(z2)]:** [2, 2]
>
> **y=2h1−h2+1:** 3

> **Main model input x:** [1, 3]

z1=x1−x2: -2

z2=x1+x2−2: 2

h=[ReLU(z1), ReLU(z2)]: [0, 2]

y=2h1−h2+1: -1

Main model input x: [1, 1]

z1=x1−x2: 0

z2=x1+x2−2: 0

h=[ReLU(z1), ReLU(z2)]: [0, 0]

y=2h1−h2+1: 1

Main model input x: [3, 0]

z1=x1−x2: 3

z2=x1+x2−2: 1

h=[ReLU(z1), ReLU(z2)]: [3, 1]

y=2h1−h2+1: 6

Main model input x: [0, 1]

z1=x1−x2: -1

z2=x1+x2−2: -1

h=[ReLU(z1), ReLU(z2)]: [0, 0]

y=2h1−h2+1: 1

Main model input x: [0, 2]

z1=x1−x2: -2

z2=x1+x2−2: 0

h=[ReLU(z1), ReLU(z2)]: [0, 0]

y=2h1−h2+1: 1

Guided: x = [1, 1] gives z = [0, 0], h = [0, 0], y = 1. Using x directly would give 2, skipping the hidden transformation.

Exit: compatible shapes and execution check whether the specified computation can run, not whether **prediction**s match real labels. Accuracy requires appropriate labelled examples, a metric and independent evaluation.

Complete separate pictured model and scope

Compose two hidden detectors before one **output** instead of skipping a **layer**.

For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

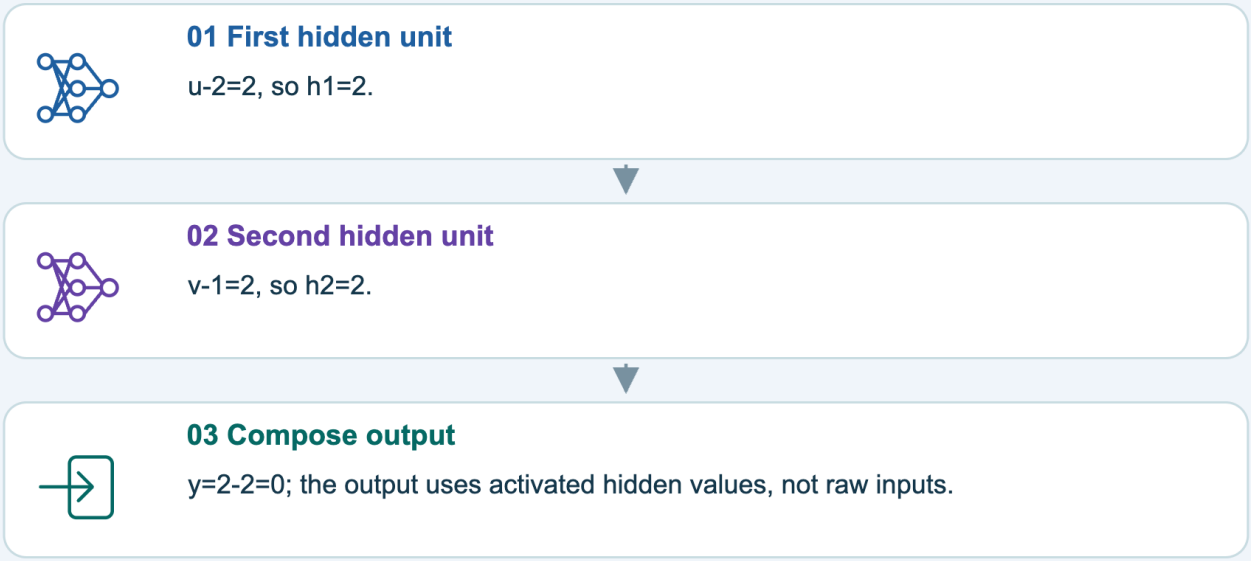
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The pictured model is a separate simpler network:h1=ReLU(u−2), h2=ReLU(v−1), y=h1−h2. At[4, 3], preactivations[2, 2]give h[2, 2], y0. Change only v3→1:preactivations[2, 0], h[2, 0], y2. Do not use the main model’s formulas or output bias 1 here. The shared idea is that the output receives the computed hidden values, not the original input. Supplied detector names do not imply a learned biological mechanism.

Trace activated values into the final difference

AI ACADEMY / AI-5/5 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. First hidden unit	u-2=2, so h1=2.
2. Second hidden unit	v-1=2, so h2=2.
3. Compose output	y=2-2=0; the output uses activated hidden values, not raw inputs.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Change: Change only v from 3 to1.

Prediction: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Layer	A group of computations that maps an input vector to an output vector
Hidden activation	An intermediate value computed inside a network
Composition	Applying one function to the output of another
Forward calculation	Computing successive values from an input using fixed parameters

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Complete each stage before using its output

AVAILABLE INPUT	COMPUTATION	VALUE PASSED ON
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

Trace x = [1, 1]. Record z1, z2, h1, h2 and y. Explain why using x directly in the output formula would be a different model.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Composing layers

The facts we will use

AI ACADEMY / AI-5/5

Composing layers

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. First hidden unit

u-2=2, so h1=2.



2. Second hidden unit

v-1=2, so h2=2.



3. Compose output

y=2-2=0; the output uses activated hidden values, not raw inputs.

CHANGE ONE THING

Change only v from 3 to1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Teacher check: Teacher explanation: Three hidden values need three matching weights under this dense **output** definition. Specify all three coefficients and the bias or explicitly design a documented selection operation.

Do not silently drop the third value.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

h2 changes from 2 to 0 because ReLU(1-1)=0; subtracting 0 rather than 2 raises y from 0 to2. Changed-case check: h1 stays 2, h2 becomes 0, y becomes 2.

The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

Changed-case reasoning: h1 stays 2, h2 becomes 0, y becomes 2. The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

Collect each learner’s explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a five-column trace table labelled x, z, h, **output** contributions and y. Allow arithmetic support while requiring the student to name each connection.

Extension: replace hidden ReLU with identity and simplify the lesson network to y = x1 - 3x2 + 3. Test [1, 3] to show that the simplified affine model differs from the ReLU model.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve ReLU at negative, zero and positive scores. Ask students to distinguish an activation from a class.
7 to 18 minutes	Write both first-layer formulas side by side. Draw arrows from the same x to both; label the ordered hidden output.
18 to 32 minutes	Trace [3, 1] together, then use W1. Require scores before ReLU and output contributions after ReLU.
32 to 45 minutes	Run or trace the optional Python and complete the guided [1, 1] case. Ask why directly substituting x into the output formula is incorrect.
Session 2 start to 8 minutes	Retrieve the two layer input/output sizes. Let students describe composition in one sentence.
8 to 20 minutes	Complete W2 and W3. Compare a swap-insensitive case with a revealing case before discussing bug coverage.
20 to 35 minutes	Collect W4 independently before hints. Check both pre-activation scores, both activations, q and class under the declared threshold.
35 to 45 minutes	Complete W5 and the exit response. Give targeted feedback and the fresh retry without replacing the original assessment record.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

W2 three hidden values need three matched coefficients and declared bias or documented selector. Main traces/probes supplied; do not use equal hidden values as the only order-bug test. Independent[2, 1]:scores 0/1, hidden[0, 1], q1, class 1; [1, 2]:scores 0/−1, hidden[0, 0], q−1, class 0. Hidden size 2, output size 1. Hidden zero is an activation, not an automatic class. Main collision[0, 1]/[0, 2]bothh[0, 0], y1.

Diagnose the current misconceptions

The second hidden unit receives the first hidden unit **output**: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Two numbers mean the connection must be correct: Matching lengths do not establish matching order or meaning. Trace a case with unequal **hidden activation**s.

A hidden unit is already a class **prediction**: Its activation is an intermediate numerical feature. A declared **output** rule assigns the final class.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

Teacher explanation: For [2, 1], scores 0 and 1 give [a, b] = [0, 1], q = 1 and class 1. For [1, 2], scores 0 and -1 give [0, 0], q = -1 and class 0. Hidden dimension is 2; **output** dimension is 1.

The equality at hidden score zero produces activation zero, not an automatic final class.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the second worked case

For x = [1, 3], write both scores, both ReLU activations and y using the lesson model.

Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

W2 Repair a dimension mismatch

A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Teacher explanation: Three hidden values need three matching weights under this dense **output** definition. Specify all three coefficients and the bias or explicitly design a documented selection operation.

Do not silently drop the third value.

W3 Use a revealing order check

Compare hidden vectors for x = [4, 1] and x = [3, 0]. Which **input** reveals a bug that swaps h1 and h2 but keeps **output** weights unchanged? Give correct and swapped **output**s for that **input**.

Teacher explanation: The first gives [3, 3] and cannot expose a swap. The second gives [3, 1]; correct **output** 6 becomes 0 when values alone are swapped.

W5 Explain information loss

Use the lesson model to verify h and y for [0, 1] and [0, 2]. Could a fixed later **layer** receiving only h distinguish the two original **input**s?

Teacher explanation: Scores are [-1, -1] and [-2, 0]. Both produce h = [0, 0] and y = 1.

A deterministic later computation given the same h produces the same result; additional original-**input** information would require a changed architecture.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: x = [1, 1] gives z = [0, 0], h = [0, 0], y = 1. Using x directly would give 2, skipping the hidden transformation.

Exit: compatible shapes and execution check whether the specified computation can run, not whether **prediction**s match real labels. Accuracy requires appropriate labelled examples, a metric and independent evaluation.

W1 Trace the second worked case

W2 Repair a dimension mismatch

W3 Use a revealing order check

W4 Compose a new network independently

A new model takes [u, v]. Compute a = ReLU(u + v - 3), b = ReLU(u - v), then q = a + 2b - 1. Trace both **input**s [2, 1] and [1, 2], including scores before ReLU. Apply class 1 when q >= 0.

State the hidden and **output** vector sizes.

Teacher explanation: For [2, 1], scores 0 and 1 give [a, b] = [0, 1], q = 1 and class 1. For [1, 2], scores 0 and -1 give [0, 0], q = -1 and class 0. Hidden dimension is 2; **output** dimension is 1.

The equality at hidden score zero produces activation zero, not an automatic final class.

W5 Explain information loss

Independent assessment and fresh retry

W4 requires both complete traces, correct classes and sizes. Fresh retry: a = ReLU(u - 1), b = ReLU(v - 2), q = 2a - b.

Inputs [0, 3] and [2, 2] give pre-activation scores [-1, 1] and [1, 0], hidden vectors [0, 1] and [1, 0], and q values -1 and 2. Under q >= 0 their classes are 0 and 1. Ask which vector the **output layer** actually receives.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled composing **layer**s **evidence** sheet to Image-pattern Network Investigation. Show how you can trace how one **layer** supplies the next **layer input**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a five-column trace table labelled x, z, h, **output** contributions and y. Allow arithmetic support while requiring the student to name each connection.

Extension: replace hidden ReLU with identity and simplify the lesson network to y = x1 - 3x2 + 3. Test [1, 3] to show that the simplified affine model differs from the ReLU model.

Week 6 develops forward passes by hand and systematic checking of intermediate values.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/6, ai-5/8, ai-5/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	

Record	Teacher evidence
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Trace how one **layer** supplies the next **layer input**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Session:** Session 3 · Investigate and transfer · 45 min
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: A revised hidden **layer** returns [h1, h2, h3]. The **output layer** still has weights [2, -1]. Explain the mismatch and what must be specified before calculation.

Reasoned practice answer: Teacher explanation: Three hidden values need three matching weights under this dense **output** definition.

Specify all three coefficients and the bias or explicitly design a documented selection operation. Do not silently drop the third value.

Check every learner before discussion: Use the worked case for Composing **layer**s. Proposed explanation: “The second hidden unit receives the first hidden unit **output**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Both units in this hidden **layer** receive x. The next **layer** receives the pair of activations after both are computed.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change only v from 3 to1. Require a **prediction** and a reason before comparing. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

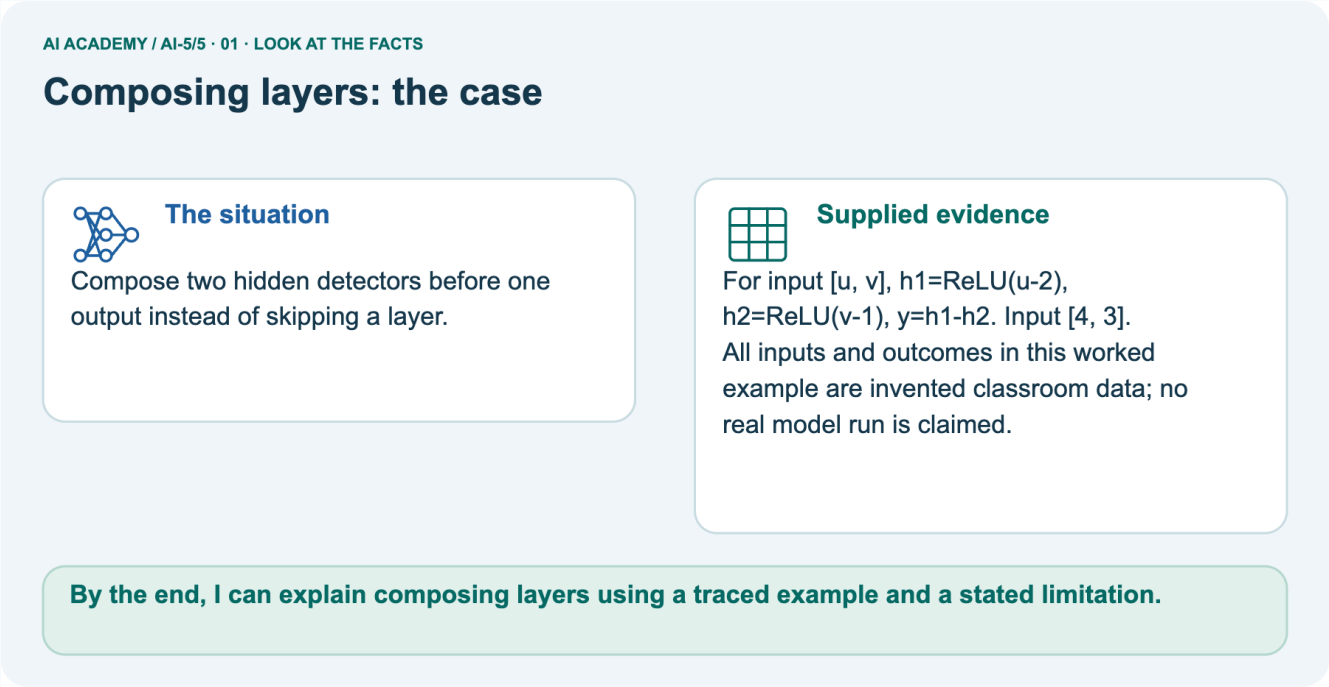
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Composing layers: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

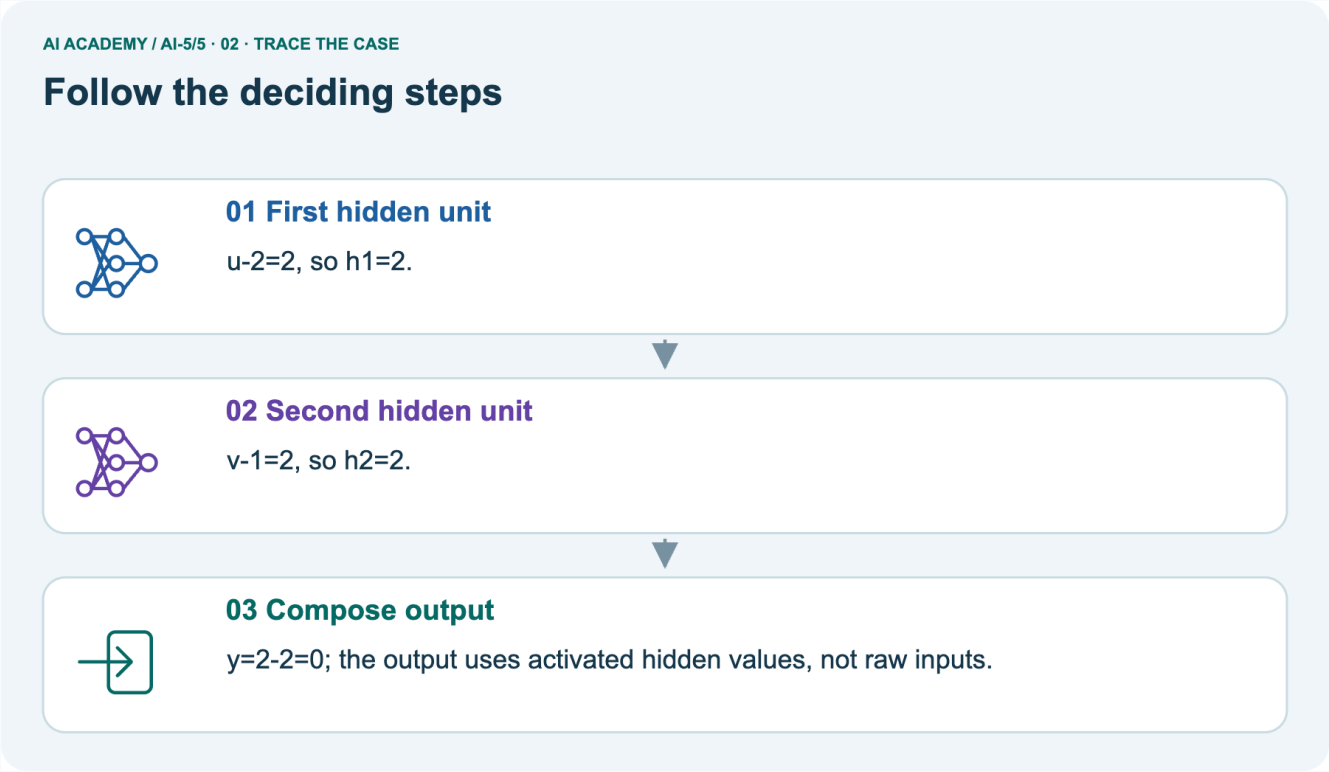
Compose two hidden detectors before one **output** instead of skipping a **layer**. Supplied facts: For **input** [u, v], h1=ReLU(u-2), h2=ReLU(v-1), y=h1-h2. **Input** [4, 3].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



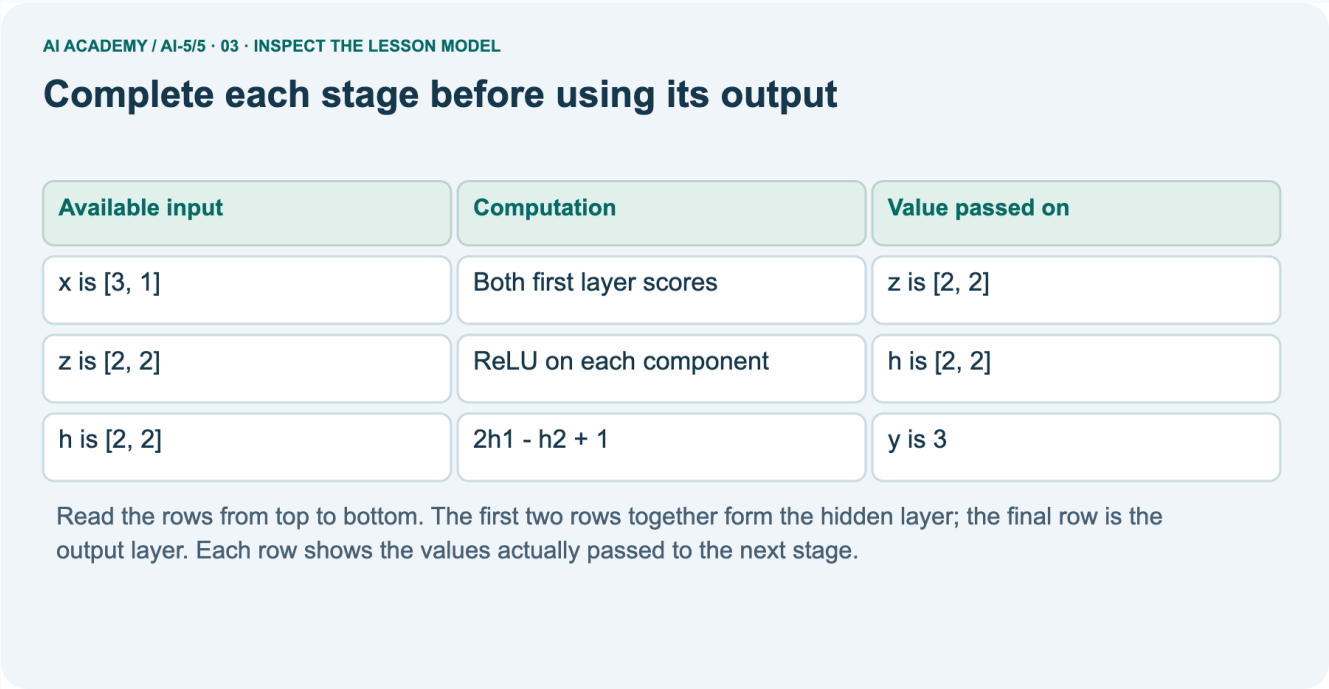
Follow the labelled picture with the matching explanation. Open the vector diagram

First hidden unit: u-2=2, so h1=2. → Second hidden unit: v-1=2, so h2=2. → Compose **output**: y=2-2=0; the **output** uses activated hidden values, not raw **input**s.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Complete each stage before using its output



AI ACADEMY / AI-5/5 · 03 · INSPECT THE LESSON MODEL

Complete each stage before using its output

Available input	Computation	Value passed on
x is [3, 1]	Both first layer scores	z is [2, 2]
z is [2, 2]	ReLU on each component	h is [2, 2]
h is [2, 2]	2h1 - h2 + 1	y is 3

Read the rows from top to bottom. The first two rows together form the hidden layer; the final row is the output layer. Each row shows the values actually passed to the next stage.

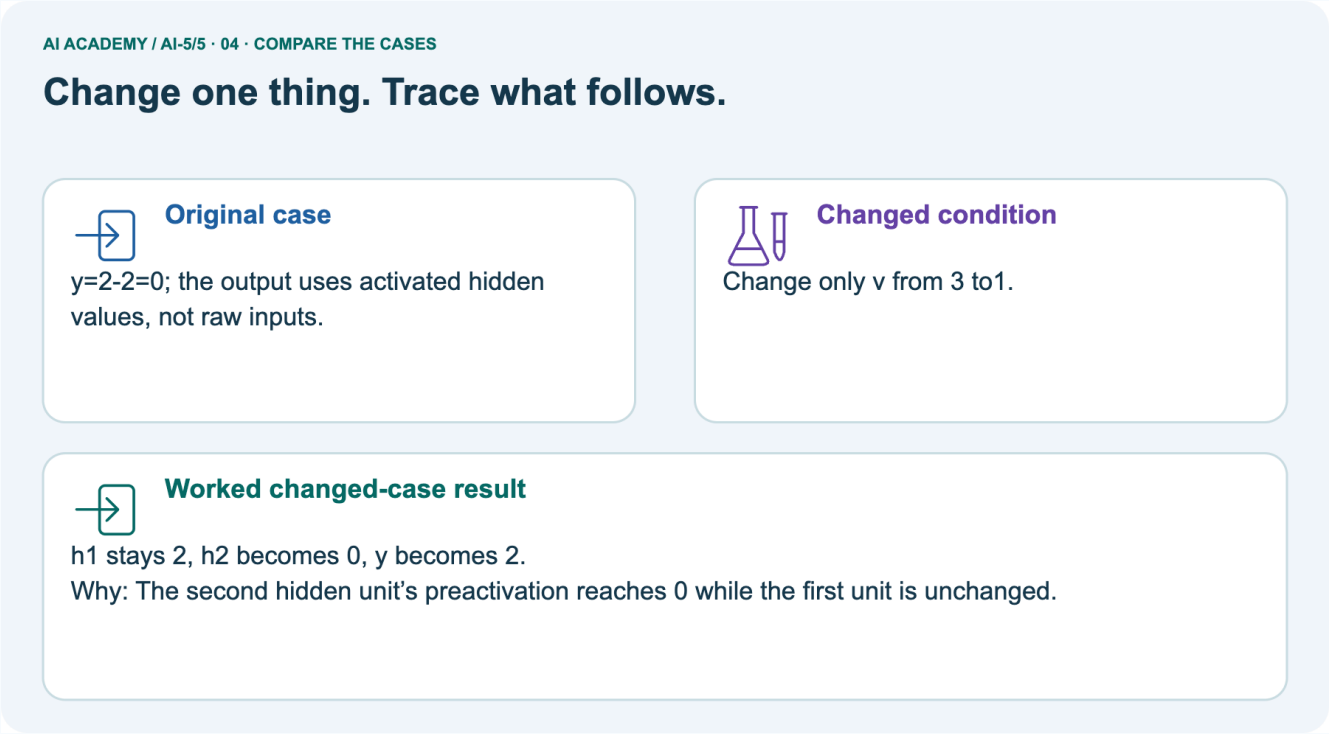
Follow the labelled picture with the matching explanation. Open the vector diagram

Complete each stage before using its **output**. x is [3, 1] | Both first **layer** scores | z is [2, 2]. z is [2, 2] | ReLU on each component | h is [2, 2]. h is [2, 2] | 2h1 - h2 + 1 | y is 3

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: y=2-2=0; the **output** uses activated hidden values, not raw **input**s. Changed condition: Change only v from 3 to1. Expected result: h1 stays 2, h2 becomes 0, y becomes 2.

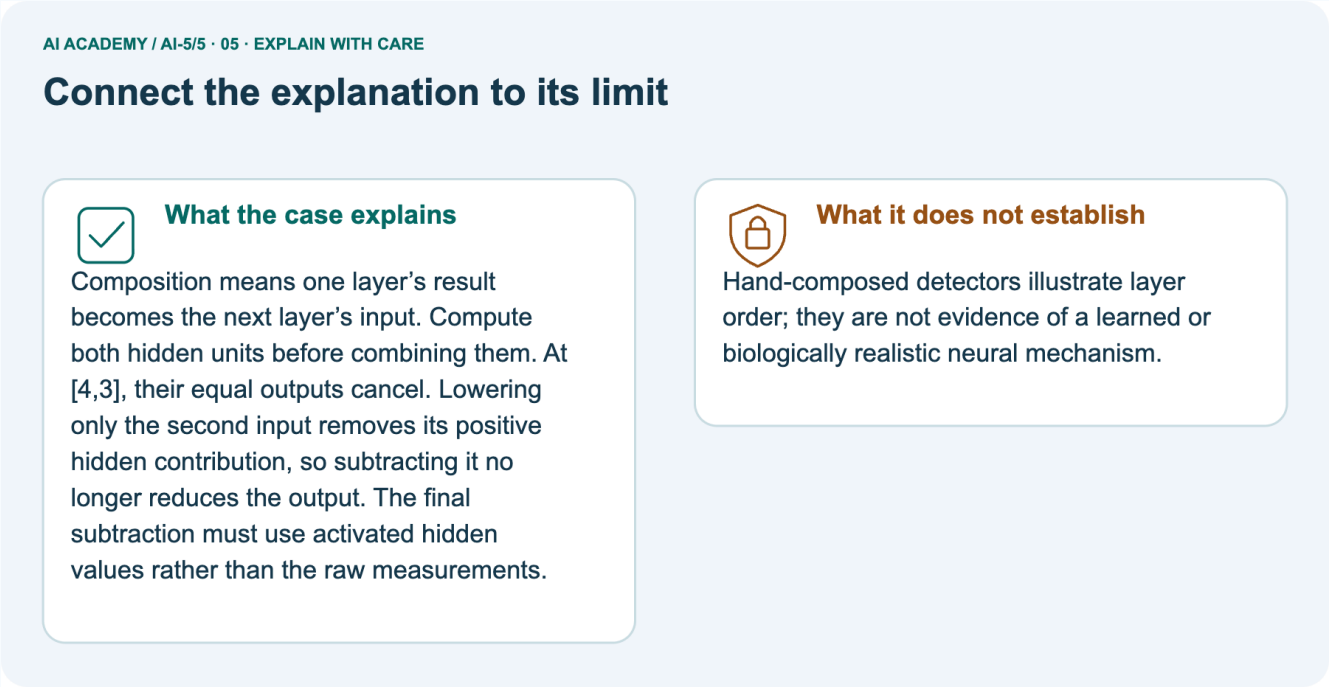
Reason: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The second hidden unit's preactivation reaches 0 while the first unit is unchanged.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Composition means one **layer**'s result becomes the next **layer**'s **input**. Compute both hidden units before combining them. At [4,3], their equal **output**s cancel.

Lowering only the second **input** removes its positive hidden contribution, so subtracting it no longer reduces the **output**. The final subtraction must use activated hidden values rather than the raw measurements.

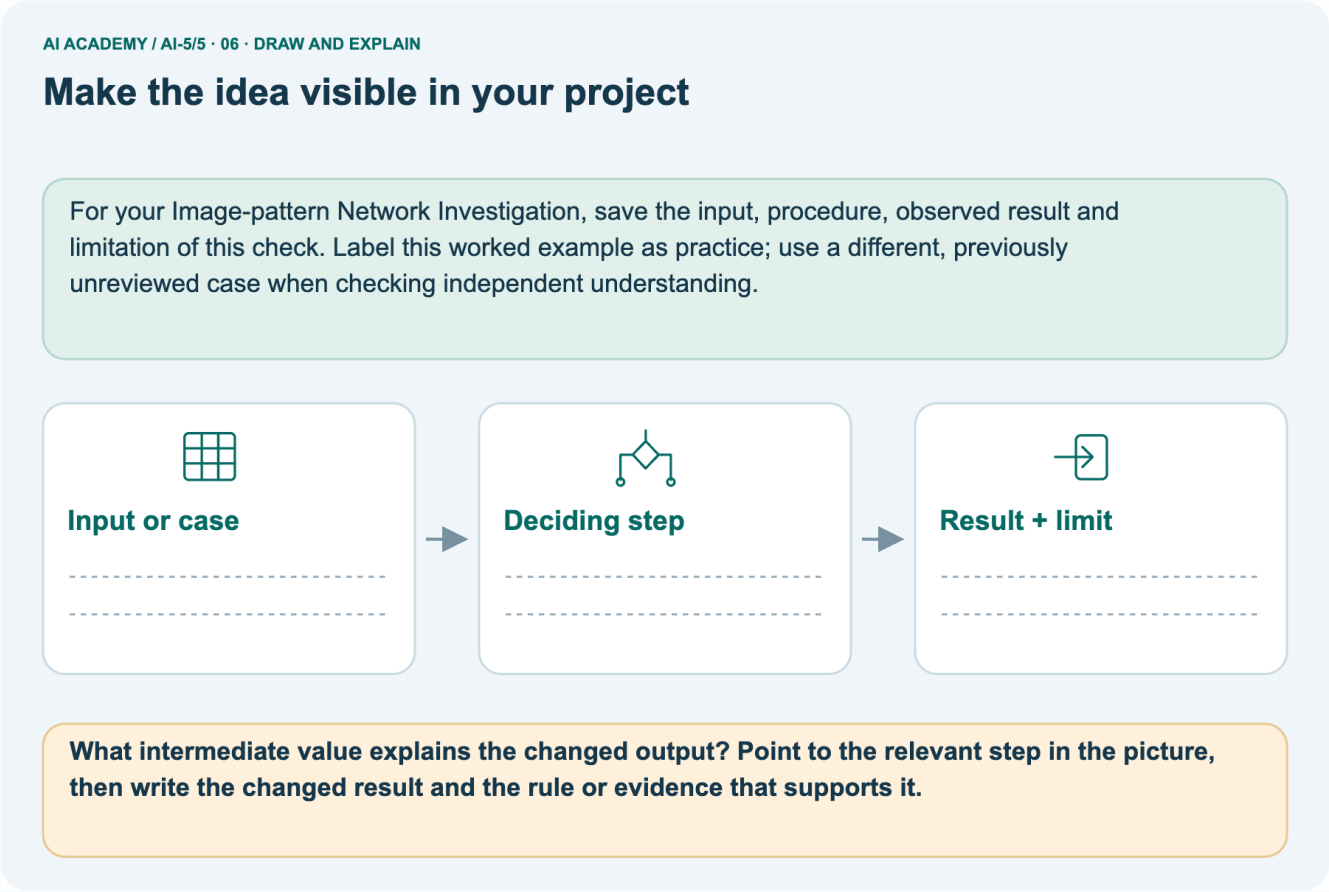
Limit: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Hand-composed detectors illustrate **layer** order; they are not **evidence** of a learned or biologically realistic neural **mechanism**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What intermediate value explains the changed **output**? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: h2 changes from 2 to 0 because ReLU(1-1)=0; subtracting 0 rather than 2 raises y from 0 to2. Changed-case check: h1 stays 2, h2 becomes 0, y becomes 2.

The second hidden unit’s preactivation reaches 0 while the first unit is unchanged.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Composing layers: the case

02. Follow the deciding steps

03. Complete each stage before using its output

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.